

Table S1 Levels of overall trends for a) trial and b) climatic conditions (average April – October) for years 1987 and 2023 and difference between levels 2023 and 1987 expressed in absolute (Diff) and relative (%) values based on level 1987 (%) (Eq. (4)).

a)

Traits	Grain maize						Forage maize					
	Maturity	1987	2023	Diff	%	sig	1987	2023	Diff	%	sig	
Days from first of January to day of sowing	early	118.1	116.2	-2.0	-1.7	ns	119.1	118.2	-1.0	-0.8	ns	q
	medium	118.3	115.9	-2.5	-2.1	ns	118.6	116.2	-2.4	-2.0	ns	
	late	118.1	116.2	-2.0	-1.7	ns	118.4	115.0	-3.4	-2.9	*	
Days from first of January to day of harvest	early	290.0	288.4	-1.6	-0.6	ns	268.5	258.9	-9.6	-3.6	ns	
	medium	289.7	285.9	-3.9	-1.3	ns	269.5	257.1	-12.4	-4.6	*	
	late	289.0	285.8	-3.2	-1.1	ns	266.0	253.1	-12.9	-4.9	*	
Days from sowing to harvest	early	171.6	172.5	0.9	0.5	ns	149.1	140.8	-8.4	-5.6	ns	
	medium	170.9	170.2	-0.7	-0.4	ns	150.9	140.7	-10.2	-6.7	ns	
	late	170.5	169.9	-0.5	-0.3	ns	147.4	138.6	-8.8	-6.0	ns	
Soil fertility points	early	59.7	63.4	3.7	6.2	***	54.8	58.7	4.0	7.2	***	q
	medium	60.6	66.4	5.9	9.7	***	56.8	61.4	4.7	8.2	***	q
	late	61.2	66.7	5.5	8.9	***	61.8	67.3	5.5	8.9	***	q

b)

Climate (average April – December)	1987	2023	Diff	%	sig	
Average yearly precipitation mm	457.4	426.8	-30.6	-6.7	ns	
Average yearly temperature °C	13.2	14.7	1.5	11.5	***	
Cumulative yearly sunshine hours h	1247.1	1443.2	196.2	15.7	**	
CO2 concentration in atmosphere ppm	349.8	422.1	72.3	20.7	***	q

q Quadratic regression function

Sign significance level; * Significant at 5% level; ** Significant at 1% level; *** Significant at 0.1% level;